

MODELING — HOW RETRIEVAL SYSTEMS THINK

The Taxonomy of Information Retrieval Models

The different philosophies a system can use to decide what counts as a match

What We'll Cover



One Big Question

Every IR model is really just a different answer to: how do we decide if a document matches?



Classical Models

Boolean, Vector Space, and Probabilistic — the three founding philosophies of IR.



Non-Classical Models

Models built on reasoning and inference, not just similarity or probability.



Alternative Models

Classical models enhanced with ideas borrowed from other fields — fuzzy logic, clustering, latent meaning.

The One Big Question Behind Every Model



Picture a librarian standing at the counter with your request in hand and a book in the other.

How do they decide whether this book counts as a match?

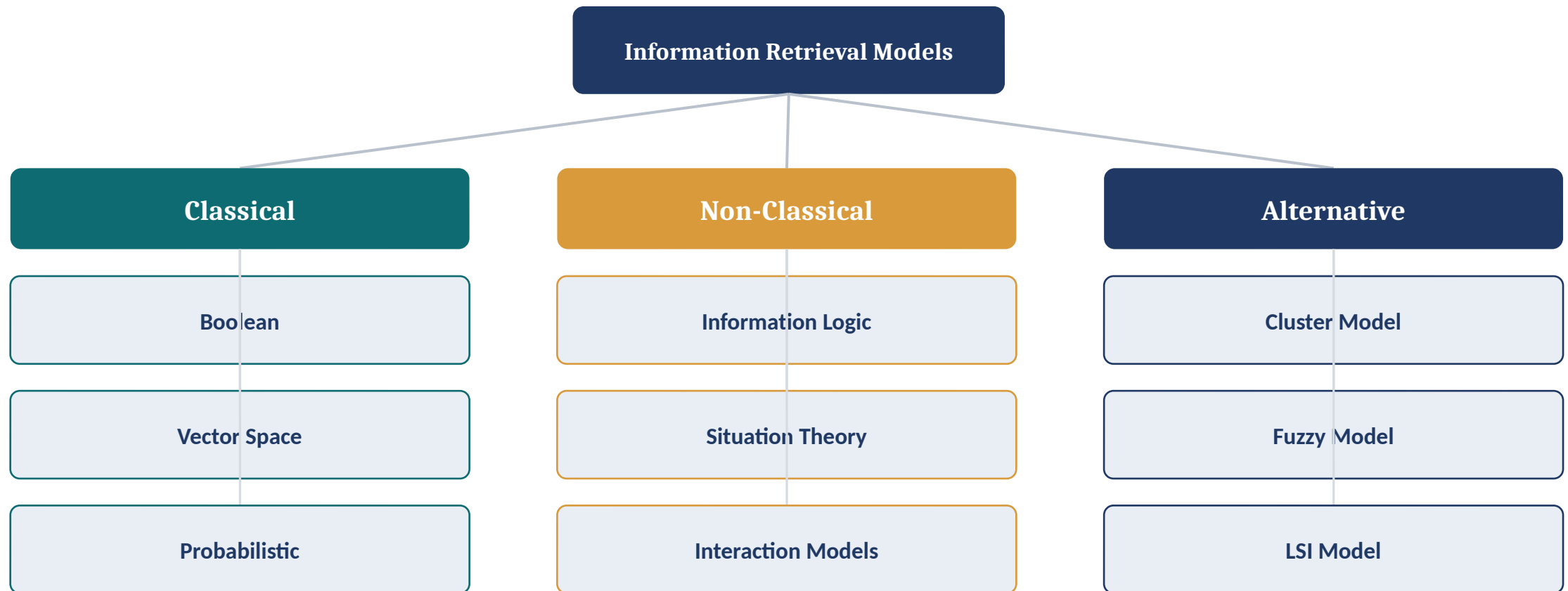
That single decision — how to judge a match — is the entire subject of “modeling” in IR. Every model we study today is just a different philosophy for making that one judgment.



Three honest ways to judge a match

- “Does it exactly satisfy my rule?” (Boolean)
- “How similar is it to what I asked for?” (Vector Space)
- “How likely is it to be relevant to me?” (Probabilistic)

The Family Tree of IR Models



Classical = the three founding philosophies. Non-Classical = reasoning-based, not similarity-based. Alternative = classical models enhanced with outside ideas.

Classical Models — The Founding Three

Every one of these treats a document as a bag of index terms, and differs only in HOW it compares that bag to your query.



Boolean

Exact, rule-based match. Yes or no — nothing in between.



Vector Space

Similarity-based match. A closeness score, ranked best-first.



Probabilistic

Likelihood-based match. Ranked by estimated probability of relevance.

The Boolean Model



The strict librarian

Imagine a librarian who only accepts a rulebook: “bring me a book with Brutus AND Caesar, but NOT Calpurnia.” They check it word for word. It either satisfies the rule, or it's rejected. No partial credit, ever.

Basis: set theory and Boolean algebra (AND, OR, NOT). A document is represented as a set of terms it contains — nothing more.

Real-world 2026 example

- Advanced search filters — site:, filetype:, exact-phrase "..." on Google.
- SQL-style legal and patent databases where a missed document could cost a case.
- Email filters: “from X AND subject contains Y.”

Strength: precise and predictable. Weakness: no ranking, and it's brittle — miss one exact word and a perfectly relevant document is rejected outright.

The Vector Space Model



The librarian who measures closeness

Instead of a strict rulebook, this librarian turns your request AND every book into a “profile” — a list of important keywords, each weighted by how strongly it stands out. Then they measure how CLOSE two profiles are, and simply rank books from closest to farthest.

Basis

Documents and queries become vectors of term weights (commonly tf-idf). Relevance is measured by the angle/distance between two vectors — closer vectors score higher, and results come back as a RANKED LIST, not a yes/no set.

Real-world 2026 example

Plagiarism checkers and “similar articles” recommendations both work by measuring how close two documents' keyword profiles are — the same core idea as vector space scoring.

The Probabilistic Model



The librarian who estimates the odds

Think of a doctor diagnosing from symptoms: not “this disease is present or absent” but “given these symptoms, how **LIKELY** is this diagnosis?” This librarian does the same thing with books — ranks every book by the estimated **PROBABILITY** that it's relevant to you, based on evidence.

The Probability Ranking Principle

“If a system ranks documents in order of decreasing probability of relevance, its overall effectiveness will be the best obtainable from the available data.” — the core rule (Manning et al., Ch. 11, after van Rijsbergen 1979)

Real-world 2026 example

Spam filters and search engines like Elasticsearch (using BM25, a modern probabilistic scoring formula) both rank by estimated likelihood, not by exact rules or plain similarity.

The Three Classical Models, Side by Side

	Boolean	Vector Space	Probabilistic
Judges match by	Exact rule (AND/OR/NOT)	Similarity / closeness	Estimated probability of relevance
Output	Unordered set	Ranked list	Ranked list
Math basis	Set theory & logic	Linear algebra (vectors)	Probability theory
Real-world use	Legal/patent search, advanced filters	Similarity & recommendation engines	Spam filters, BM25-based search

Non-Classical Models — Reasoning, Not Just Matching

All three classical models share one hidden assumption: relevance is a similarity or probability SCORE, computed once, from a fixed query and a fixed document. Non-classical models drop that assumption and treat retrieval as a process of reasoning or inference instead.



Information Logic Model

Treats a document as a set of logical propositions. A document is relevant if it logically IMPLIES the query — closer to a mathematical proof than a similarity score.



Situation Theory Model

Relevance depends on the situation and context surrounding the information need, not just the words — the same query can mean different things in different contexts.



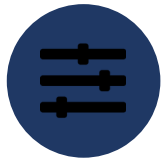
Interaction Model

Relevance is negotiated through back-and-forth interaction between user and system — closer to a conversation than a one-shot lookup.

Alternative Models — Enhancing the Classics



These models don't reject the classical philosophies — they keep the same foundation (usually set theory or vector space) and bolt on ONE specific idea borrowed from another field to fix a specific weakness.



Fuzzy Model

Borrows fuzzy logic: relevance isn't 0 or 1, it's a graded degree in between.



Cluster Model

Borrows clustering: group similar documents first, then search within the right group.



LSI Model

Borrows linear algebra: finds hidden, “latent” connections between related words.

The Fuzzy Model



The librarian who allows “sort of”

Classical set theory says a term is either IN a document or NOT — a hard 0 or 1. Fuzzy set theory allows a document to belong to a query's “relevant set” by a DEGREE — 0.3 relevant, 0.8 relevant — instead of forcing a binary decision.

Real-world 2026 example

Streaming platforms like Netflix or Spotify recommending something “similar, but not identical” to what you just watched or heard — that soft, graded sense of match is exactly the fuzzy idea: not a strict yes/no, but a dial of how well something fits.

Why it helps: it softens the Boolean model's harsh all-or-nothing cutoff, without needing the full machinery of vector space or probability.

The Cluster Model



The librarian who organizes by genre first

Instead of comparing your request to every single book in the entire library one by one, this librarian first groups similar books together into sections — mystery, cooking, engineering — and only then searches within the section your request belongs to.

Real-world 2026 example

Google News grouping dozens of articles about the same event into a single story cluster, instead of listing every near-identical article separately — the same underlying idea as clustering documents before retrieval.

Why it helps: searching within a small, relevant group is faster and often more accurate than comparing against the entire collection every time.

The LSI Model (Latent Semantic Indexing)



The librarian who reads between the lines

Every model so far compares the literal WORDS used. LSI notices something deeper: “car” and “automobile” almost never appear in the exact same sentence, yet they clearly mean related things, because they show up in similar surrounding company across many documents. LSI mathematically uncovers this hidden (“latent”) relationship.

Real-world 2026 example — this is very current

Modern “semantic search” — where searching “buy running shoes” also returns pages about “purchase athletic footwear” — descends directly from this same latent-meaning idea. LSI was an early version of what today's AI-powered search calls embeddings.

Why it helps: it solves the exact problem the pipeline-leaks example from Unit 1 raised — matching MEANING, not just matching letters.

All Nine Models, One Table

Branch	Model	Judges relevance by
Classical	Boolean	Exact logical rule match
Classical	Vector Space	Similarity between weighted term vectors
Classical	Probabilistic	Estimated probability of relevance
Non-Classical	Information Logic	Logical implication between document and query
Non-Classical	Situation Theory	Context/situation surrounding the need
Non-Classical	Interaction Model	Ongoing negotiation between user and system
Alternative	Fuzzy Model	Graded (partial) degree of match
Alternative	Cluster Model	Membership in a pre-grouped cluster of similar documents
Alternative	LSI Model	Latent (hidden) semantic similarity, beyond literal words

Key Terms — One-Glance Recap

Model (in IR)

A mathematical philosophy for deciding whether/how well a document matches a query.

Boolean Model

Exact match using AND/OR/NOT; output is an unordered set.

Vector Space Model

Documents & queries as weighted term vectors; ranked by similarity.

Probabilistic Model

Ranks by estimated probability of relevance (Probability Ranking Principle).

Non-Classical Models

Reasoning/inference-based models — logic, context, or interaction, not just similarity.

Alternative Models

Classical models enhanced with an outside technique (fuzzy logic, clustering, latent semantics).

Fuzzy Model

Allows a graded, partial degree of relevance instead of strict yes/no.

Cluster Model

Groups similar documents first, then searches within the relevant group.

LSI Model

Finds hidden semantic relationships between words that rarely co-occur literally.

One Sentence to Remember

Every IR model is just a different, honest answer to one question — how do we decide a document matches — and the taxonomy simply groups those answers by the KIND of reasoning they use: exact rules, similarity, probability, logic, context, or a borrowed enhancement.

Exam pointers for this topic

- Draw the full family tree (Classical / Non-Classical / Alternative) from memory.
- Differentiate Boolean, Vector Space, and Probabilistic models with one line each.
- Explain why Alternative models are “enhancements” rather than new philosophies.
- Give one real-world example for at least three different models.